

Spring 2026
Introduction to Philosophy of Technology
Mid-Semester Examination

28th February, 2026

Time: 1.5 hours
Marks: 40

Answer any 4 in about 400 - 500 words; keep your answers precise and to the point

4 × 10

1. What according to Jathan Sadowski do the metaphorical role models of (a) the mechanic and (b) the Luddite bring to the materialist analysis of technological capitalism? What argument does the author provide for the necessity of embodying both the models (the mechanic and the Luddite)? How is the model of the ‘mechanic’ different from that of the ‘engineer’ in the author’s account?

(a) Mechanic as a model for analysis of technological capitalism

What does it mean to be a mechanic? (1 mark; 0.5 for each valid point; look for any two)

- Being a mechanic means having a trade, a profession, a certification, a degree, a set of skills.
- being a mechanic is as simple as pursuing a curiosity about how the world really works and what you can do in it.

What it means for analysis of technological capitalism? (1 mark; 0.5 for each valid point; look for any two)

- be empowered to get under the hood and tinker with the machines, as well as take apart and (re)build the machines for our own purposes.
- set us on the pathway to a more subversive techno-politics.
- It is not enough to simply “fix” the systems that already exist—in the sense of restoring and maintaining the machinery of capital—we must also scrap and salvage, take apart and build anew.

(b) Luddite as a model for analysis of technological capitalism

What does it mean to be a luddite? (1 mark; 0.5 for each valid point; look for any two)

- being a Luddite is opening yourself to the radical possibility of evaluating the technologies that fill our lives and determining whose goals they primarily advance
- Luddism is about asserting the ability to make meaningful decisions about the directions and applications of technologies that structure our lives.

What it means for analysis of technological capitalism (1 mark; 0.5 for each valid point; look for any two)

- to perform critical analysis of the roles, purposes, and impacts of technology in society
- to know why the machine was built, whose purposes it serves, and when it should be seized.

Argument for embodying both the models (1 mark, 0.5 for each; look for at least 2 points)

- The author argues for how we should always strive to embody both of these models. Neither is sufficient on its own.

- The materialist analysis of the mechanic provides the basis for the material action of the Luddite.

Mechanic vs. engineer (look for any 2 points; 1 mark each)

- The author notes as to how as the tools and machines became more complex and as both industrialists and intellectuals started granting technology the power to drive and determine the shape of modern society, the mechanic was supplanted by the engineer, who was meant to serve the imperatives of capital
 - Whereas the term **mechanic** (or industrial, or practical) arts calls to mind men with soiled hands tinkering at workbenches, **technology** conjures clean, well-educated, white male technicians in control booths watching dials, instrument panels, or computer monitors.
 - Whereas the mechanic arts belong to the mundane world of work, physicality, and practicality—of humdrum handicrafts and artisanal skills—technology belongs on the higher social and intellectual plane of book learning, scientific research, and the university.
 - There is a **gendered dimension to engineering**. As soon as work once classified as menial, mundane, and mindless becomes especially valuable for capitalist exploitation, then suddenly it gets redefined as a high-status, high-skill, high-tech profession. The social category shifts from **feminine to masculine, from racialized to white, from poor to prestigious**.
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2. How has the effect of social relations on technical design been documented by historians studying the politics of design in technologies? Provide any 2 instances of such accounts. What do these analyses reveal about the relation between technology and society?

Effect of social relations on technical design (1 mark each; look for any 2 points; 2 marks)

- By documenting the **contingency of design**, identifying in stances where "**things could have been different**," where, for example, the same artifact could have been made in different ways, or differently designed artifacts could have been constructed.
- Having identified contingency, the historian can then ask **why one way, or one design, was chosen rather than another**. In that way the question of the effect of social relations becomes a matter for empirical inquiry as well as for theory.

Any 2 instances (any two from below; 2.5 marks for each instance; 5 marks total)

- **Langdon Winner's stimulating essay "Do Artifacts Have Politics?"** provides a rudimentary but clear example. Designing the bridges over Long Island parkways, builder Robert Moses could have had them constructed with a wide range of clearances. He chose to build them low, with "as little as nine feet of clearance at the curb." The reason was that the buses which might otherwise take poor people and blacks along the parkways to Moses's "widely acclaimed public park" at Jones Beach were 12 feet high ! 88 (Why contingency is important is obvious here. If it had not been clearly possible for Moses to choose to build higher overpasses, we would have no way of assessing the relevance of his social prejudices to his bridge design.)
- One piece of work that can be seen as attempting to apply this perspective to technical design is **David Noble's analysis of the automation of machine tools**. Noble identifies contingency in that development. There were two ways to automate-record-playback and

numerical control-and it is far from clear that only numerical control was a priori viable. He also identifies a problem of valorization : the capacity of skilled machinists to control the pace of production, or indeed to disrupt it completely. He suggests that the choice of numerical control reflected its perceived superiority as a solution to this problem of valorization.

- There are many people now working to reveal the extent of contingency by designing "alternative technologies." Best known are the attempts to embody in technology the virtues of small scale, decentralization, and ecological awareness. But there are also attempts from within high-technology industry to alter in fundamental ways both what is produced and how it is produced. In Britain, this is best exemplified by the **"alternative plans" put forward by the work force at Lucas Aerospace**. These involve attempts to shift production from military to "socially useful" products, and also to change the nature of production, to reverse in practice deskilling and the separation of head and hand.

What they reveal about relation between technology and society (look for any 2 points; 1.5 marks each; 3 marks)

- By making contingency and choice actual rather than merely hypothetical, they throw into ever-sharper light **the ways in which social relations shape technical development**ⁱ. Perhaps too the process can be dialectical, rather than one-way.
- Understanding how existing technology has been and is being socially shaped **can help in reconstructing it**.
- Rescues the history of technology from being increasingly internalistic by offering a way of **combining attention to technical detail with concern for broader issues of social history**.

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3. What are the three misconceptions surrounding Luddism according to Jathan Sadowski, and how does the author clarify these? Bring out the distinction the author makes between 'politics of refusal' and 'ethics of acceptance'.

Three misconceptions surrounding Luddism and clarification of each (3 marks for each misconception; look for at least 3 points under each; 1 mark for each point; 9 marks)

- **First misconception:** the Luddites were **not indiscriminate in the machines they broke**;
 - **author's clarification**
 - They were **intentional and purposeful about which machines they smashed**. They **targeted those owned by manufacturers who were known to pay low wages, disregard workers' safety, or speed up the pace of work**.
 - Even within a single factory—which might contain machines owned by different capitalists—some machines were destroyed and others passed over depending on the business practices of their owners.
 - Thus, the machines were not their real targets; **their targets were the owners of those machines and the type of society they were bringing into being**. The Luddites were **striking blows against capitalism as a social system of exploitation by smashing the material manifestations of capital**.
- **Second misconception:** The Luddites were not ignorant;
 - Author's clarification:

- It is important to remember that the **Luddites were also mechanics**—sometimes in the literal sense and often in the figurative sense I use above—who **pushed their knowledge into action**.
- They were not mindless workers or moronic peasants; **they were skilled artisans** who knew better than most about **how the technologies and factories were operated and how labor and capital was organized**.
- Luddism was a **working-class movement opposed to the political economic consequences of industrial capitalism**. Far from being a kneejerk reaction to new technology, **smashing machines was a tactical response by workers based on their understanding of how owners were using those machines and what purposes they were serving**.
- **Third misconception:** the Luddites were not against innovation.
- **Author's clarification**
 - In fact, **many of the machines they destroyed weren't even new inventions**. One machine they targeted, the gig mill, had been used for more than a century in textile manufacturing. Similarly, wide stocking frames and power looms had been used for decades.
 - If it was fear of new things that drove the Luddites, then **they would not have waited so long to act**. If it was hatred of new innovation, then **they would not have used the machines for so long before destroying them**.
 - "Protest stemmed **not from the invention but from the application of these new technologies**," writes historian Adrian Randall.
 - The Luddites **were provoked to action when capitalists began using these machines to displace and disempower workers, make labor conditions more miserable, reduce the quality of outputs, and increase their own profits**.

Politics of refusal vs. ethics of acceptance (1 mark; 0.5 marks for clarifying each)

- the politics of refusal just means having the right and ability to say no to things that directly impact your life. Rather than a politics of refusal, we are given an **ethics of acceptance**.
 - The driving concern for the **ethics of acceptance** is: How do you get people to trust a technology, integrate it into their lives, or just look the other way and not raise a fuss? **Ethics is put in service of technology: What must we do to make you trust us and accept this thing?**
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4. What are some of the aspects of Marx's conception of technology that make him out to be a technological determinist? Outline any three points that show the limitations of reading Marx as a technological determinist.

Some aspects of Marx as a technological determinist

What is technological determinism (1 mark; any 2 points; 0.5 marks each)

- technological determinists see technology as the driving force of history. At the same time, they grant autonomy to technology – it somehow exists outside of social relations.
- Once technological innovations are introduced into society they inevitably transform it. Technology structures the social.
- Social organization is nothing more than the outcome of technological effects. The issue, then, is not one of technology and social theory but technology *as* social theory.

What makes Marx out to be a technological determinist? (any two points; 1 mark each; 2 marks)

- Marx's statement that "The handmill gives you society with the feudal lord; the steam-mill society with the industrial capitalist" – is one of the best statements of technological determinism available to us.
- Moreover, Marx's materialist conception of history may be interpreted as giving priority to the forces of production over the relations of production, with the forces of production referring to technology.

Any three points that show the limitations of reading Marx as a technological determinist (2 marks for each point; any 3; 6 marks)

- For Marx, **machines are neutral a priori**. The essence of machinery, so to speak, should not be understood as its current utilization. Marx was abundantly clear on this point: **The application of machinery in the present day is one of the relations of our present economic system, but the way in which machinery is utilised is totally distinct from the machinery itself.** Powder is powder whether used to wound a man or to dress his wounds.
- What machinery is 'is totally distinct from the machinery itself' (Marx, 1978, p. 140). **Productive forces can be harnessed for social good. This brings relations of production into focus. True oppression is social not mechanical.** The real enemy is the capitalist class. This distinction between forces and relations of production is critical. Marx (1990, pp. 554–5) gives the determining role to the relations of production, defined as the ways in which labour is socially organized.
- **Marx's critique of the Luddites:** Technology is neither society's driver nor the source of all problems; **the issue is technology's integration into society.** Technological development is distorted under capitalism because it is impelled by the logics of profit and domination. Marx (1990, pp. 554, 558–9) had observed that, for the first time, **the workers' struggle had raged against the instruments of production themselves (Luddites). The war against machines was misplaced. It confused machinery's current use for its essence. For Marx, machines are neutral a priori.**
- Marx-the-technological-determinist makes even less sense when we recall the central concern of his studies and what he took to be the true subject of history. **Marx had no interest in 'sacred' history in the Hegelian fashion.** Instead of a history of ideas he turned to 'profane' history; **the history of human beings and their concrete relations.** He did not write 'machines make history, but not in circumstances of their choosing'. **He wrote about men making their own history, he wrote about the shock of body against body, he wrote about bloody struggle, and for good reason.**
- For Engels and Marx (1956, p. 125), **the only history worthy of the name was embodied history: 'History does nothing, it possesses no immense wealth, it fights no battles. It is instead the real human being, the real living person, who does everything, who owns everything and who fights all battles'.**
- 'Technology,' Marx writes in a key footnote in Capital, **'discloses man's mode of dealing with Nature, and the process of production by which he sustains his life, and thereby also lays bare the mode of formation of his social relations, and of the mental conceptions that flow from them.'** **'Disclosing' and 'laying bare' do not denote 'determine.'**
- Marx is no technological determinist. For him, **the level of technological development is characteristic of a particular stage of the evolution of human society, not as the cause but as an indicator.** Thus manufacture precedes the era of modern industry, and capitalist relations of production are established before machinery enters the scene. **Machinery is itself an outcome of the class struggle between labour and capital and of competition among capitalists.**

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5. What is epistemological Luddism according to Langdon Winner, and what does it bring to light? In what way is Winner's proposal different from Luddism in the traditional sense? Briefly outline the underlying conception of technology in Winner's proposal for epistemological Luddism.

What is Epistemological Luddism (EL) (2 marks; 1 mark for each point; look for any 2)

- EL is the **method** of carefully and deliberately dismantling technologies with the objective of recovering the buried substance upon which our civilization rests.
 - It is crucial that they refer to it as a '**method**'!
- Once unearthed, that substance could be scrutinized, criticized, and judged.

What it brings to light (1 marks each; 3 marks)

- (1) **the kinds of human dependency and regularized behavior** centering upon specific varieties of apparatus,
- (2) **the patterns of social activity** that rationalized techniques imprint upon human relationships, and
- (3) **the shapes given everyday life by the large-scale organized networks of technology**. Far from any wild smashing, this would be a meticulous process aimed at restoring significance to the question, **What are we about?**

EL vs. Luddism in the traditional sense (2 marks; for any 2 of the below 3 points; 1 mark each)

- EL is a **method**, whereas Luddism in the traditional sense is **call to action** to break machines.
- In EL, the most interesting parts of the technological order are **not those found in the structure of physical apparatus anyway**. This is related to the point of **technologies being forms of life**.
- Luddism seen in this context would seldom refer to dismantling any piece of machinery. It would seek to **examine the connections of the human parts of modern social technology**.

Underlying conception of technology (2); 1 mark for each point

- technologies are '**forms of life**'. (they must have used this phrase!)
- They are **patterns of human consciousness and behavior adapted to a rational, productive design**.

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6. Explain Marx's materialist conception of history. What role does technology play in this analysis? Outline Marx's critique of the Luddite response to the machinery question?

Marx's materialist conception of history:

Look for (2 for covering all components of the first point; 2 for covering all components of the second; 4 marks)

- references to dialectic, relations of production, forces of production, base, superstructure.
- How the superstructure emerges from the base; How at a certain stage of their development the material forces of production in society come into conflict with the existing relations of

production. From forms of development of the forces of production these relations turn into their fetters.

Role of technology (0.5 for each point; look for any 4; 2 marks);

- Humans are labouring animals. They conscript tools as part of their labour for survival.
- Tools enable the creation of surplus. And this surplus is appropriated by a few who form the ruling class in society. The rest form the working class.
- Technology transforms society and corresponds to different stages in history, e.g., handicraft, manufacturing, and modern industry
- with the advancement of machinery, there is a tension between the forces of production and relations of production, which causes society to transform towards a new mode of production
- Technology alienates workers from their product.
- Machinery reduces the requirement for skilled labour, it creates unemployment in new areas, and its refinement undermines existing jobs
- Technology/machinery enables the capitalist to extract more surplus from the labour of the worker and exploit the worker.

What is the machinery question (1 mark)

- With momentous progress also came human cost in terms of quality of life. The machinery question had to do with how to understand the effect of technology/machinery on humanity.

Luddite response to the machinery question (1 mark)

- The Luddite response was to wage war against the machines by organizing in groups and smashing them.

Marx's critique of the Luddite response (2 marks; any 2 points; 1 mark each)

- Marx argued that the war against machines was misplaced. It confused machinery's current use for its essence. It is not machinery as such but its use by capitalists for exploitation of labour that is the root cause of the problems plaguing society. True oppression is social not mechanical. The real enemy is the capitalist class.
- Productive forces can be harnessed for social good.

